Name

Mathematics Stage 5 Core

Algebraic Fractions

Book 1

Version: 261006

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-ALG-C-01 simplifies algebraic fractions with numerical denominators and expands algebraic expressions

Apply the 4 operations to simplify algebraic fractions with numerical denominators

- Simplify expressions that involve algebraic fractions with numerical denominators

Algebraic Fraction Addition: Same Denominator

Review

1 Simplify fractions

a $\frac{6}{12}$	**b** $\frac{18}{24}$	**c** $\frac{9}{24}$	**d** $\frac{16}{16}$
e $\frac{14}{7}$	**f** $\frac{12}{18}$	**g** $\frac{8}{56}$	**h** $\frac{9}{27}$

2 Add and subtract fractions

a $\frac{5}{7}+\frac{1}{7}$

b $\frac{2}{15}+\frac{8}{15}+\frac{1}{15}$

c $\frac{6}{7}-\frac{2}{7}-\frac{4}{7}$

3 Simplify expressions by combining like terms.

a $7a+12a$

b $16u-3u$

a $3+b$

b $2x+3+5x$

c $7ab+b$

d $-4a+5a-7a$

Adding and Subtracting Algebraic Fractions

Algebraic fractions have **algebraic expressions** in their **numerator** or **denominator**.

$$\underbrace{\frac{2}{3} \quad \frac{5}{7} \quad \frac{20}{3}}_{\text{Fractions}} \qquad \underbrace{\frac{2x}{5} \quad \frac{3}{7a+4} \quad \frac{2x-4y}{7a+9b}}_{\text{Algebraic fractions}}$$

The **denominators must be the same** before adding or subtracting algebraic fractions.\

Adding fractions	Subtracting fractions
$\frac{a}{n}+\frac{b}{n}=\frac{a+b}{n}$	$\frac{a}{n}-\frac{b}{n}=\frac{a-b}{n}$

Example Add/subtract algebraic fractions with the same denominator

$\frac{5x}{13}+\frac{3x}{13}$

$=\frac{5x+3x}{13}=\frac{8x}{13}$

$\frac{5x}{13}-\frac{3y}{13}$

$=\frac{5x-3y}{13}$

Guided Practice

Simplify these expressions.

a $\frac{x}{1}+\frac{2x}{1}$ $3x$

b $\frac{x}{4}+\frac{2x}{4}$ $\frac{3x}{4}$

c $\frac{a}{3}+\frac{2a}{3}$ a

d $\frac{3y}{5}-\frac{y}{5}$ $\frac{2y}{5}$

e $\frac{4p}{6}-\frac{p}{6}$ $\frac{p}{2}$

f $\frac{2y}{15}+\frac{3y}{15}$ $\frac{y}{3}$

Key Ideas

1. When adding or subtracting fractions with the same denominator, we add or subtract the …… and keep the …… the same.

1 Add and subtract these fractions.

a $\frac{3x}{10} + \frac{4x}{10} = \frac{\square}{10}$ $\frac{7x}{10}$

b $\frac{5a}{8} - \frac{2a}{8} = \frac{\square}{\square}$ $\frac{3a}{8}$

c $\frac{4c}{7} + \frac{2c}{7} - \frac{c}{7} = \frac{\square}{\square}$ $\frac{5c}{7}$

d $\frac{4k}{3} + \frac{k}{3}$ $\frac{5k}{3}$

e $\frac{3y}{5} - \frac{y}{5}$ $\frac{2y}{5}$

f $\frac{7p}{13} - \frac{2p}{13}$ $\frac{5p}{13}$

g $\frac{10r}{7} - \frac{2r}{7}$ $\frac{8r}{7}$

h $\frac{8q}{5} - \frac{2q}{5}$ $\frac{6q}{5}$

i $\frac{8q+5}{11} + \frac{2q}{11}$ $\frac{6q+5}{11}$

j $\frac{3x}{16} + \frac{5x}{16} + \frac{7x}{16}$ $\frac{15x}{16}$

k $\frac{9e}{11} - \frac{4e}{11} - \frac{e}{11}$ $\frac{4e}{11}$

l $\frac{5}{3k} - \frac{2}{3k} + \frac{4}{3k}$ $\frac{8}{3k}$

2 Add and subtract these fractions.

a $\frac{2y}{3} + \frac{5y}{3}$ $\frac{7y}{3}$

b $\frac{11d}{7} - \frac{5d}{7}$ $\frac{6d}{7}$

c $\frac{24a}{31} - \frac{13a}{31}$ $\frac{11a}{31}$

d $\frac{8b}{19} + \frac{5b}{19}$ $\frac{13b}{19}$

e $\frac{100k}{21} - \frac{99k}{21}$ $\frac{k}{21}$

f $\frac{5g^2}{47} + \frac{12g^2}{47}$ $\frac{17g^2}{47}$

g $\frac{a}{2} - \frac{a}{2}$ 0

h $\frac{7p}{9} + \frac{p}{9}$ $\frac{8p}{9}$

3 Add and subtract these fractions, then simplify the result.

a $\frac{5x}{10} + \frac{3x}{10} = \frac{8x}{10} = \frac{4x}{5}$ (÷ 2, ÷ 2)

b $\frac{a}{8} + \frac{a}{8} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{a}{4}$

c $\frac{3n}{4} + \frac{n}{4} = \frac{\square}{\square} = __$ n

d $\frac{8}{9x} - \frac{2}{9x} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{2}{3x}$

e $\frac{3c}{2} + \frac{5c}{2} = \frac{\square}{\square} = __$ $4c$

f $\frac{11h}{18} - \frac{h}{18} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{5h}{9}$

g $\frac{11}{3x} - \frac{2}{3x} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{3}{x}$

h $\frac{5y}{12} + \frac{3y}{12} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{2y}{3}$

i $\frac{13a}{18} - \frac{4a}{18} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{a}{2}$

Algebraic Fraction Addition: Different Denominator

Review

1 Add and subtract fractions

a $\frac{5}{7}+\frac{1}{7}$

b $\frac{2}{15}+\frac{8}{15}+\frac{1}{15}$

c $\frac{6}{7}-\frac{2}{7}-\frac{4}{7}$

d $1-\frac{1}{4}$

e $1-\frac{3}{5}$

f $2-\frac{6}{7}$

2 Find the lowest common denominator (LCD) of these fractions.

a $\frac{}{3}$ and $\frac{}{5}$

b $\frac{}{4}$ and $\frac{}{5}$

c $\frac{}{7}$ and $\frac{}{6}$

d $\frac{}{3}$ and $\frac{}{6}$

e $\frac{}{3}$ and $\frac{}{5}$

f $\frac{}{4}$ and $\frac{}{5}$

g $\frac{}{12}$ and $\frac{}{6}$

h $\frac{}{10}$ and $\frac{}{20}$

3 Find the missing number to make these fractions equivalent.

a $\frac{2}{3}=\frac{\square}{6}$

b $\frac{4}{7}=\frac{\square}{21}$

c $\frac{1}{3}=\frac{\square}{12}$

d $\frac{6}{11}=\frac{\square}{55}$

e $\frac{x}{5}=\frac{\square}{10}$

f $\frac{2a}{7}=\frac{\square}{21}$

g $\frac{4z}{5}=\frac{\square}{20}$

h $\frac{3k}{10}=\frac{\square}{50}$

i $\frac{3c}{5}=\frac{\square}{10}$

j $\frac{v}{2}=\frac{\square}{16}$

k $\frac{7a}{8}=\frac{\square}{24}$

l $\frac{5n}{6}=\frac{\square}{42}$

4 Add and subtract these fractions.

a $\frac{7}{20}+\frac{5}{10}$

a $\frac{7}{10}-\frac{2}{5}$

b $\frac{11}{12}-\frac{1}{3}-\frac{1}{4}$

c $\frac{1}{7}+\frac{2}{5}$

b $\frac{1}{4}+\frac{2}{3}$

d $\frac{3}{7}-\frac{2}{9}$

Adding and Subtracting Algebraic Fractions (Different Denominator)

1. Rewrite the fractions with the **lowest common multiple** of the denominators.
2. Add or subtract the numerators.

Example Add/subtract algebraic fractions with different denominators

$\frac{4x}{3} - \frac{x}{6}$

1. Same denominators $\frac{\quad}{6} - \frac{\quad}{6}$

$\frac{8x}{6} - \frac{x}{6}$

2. Subtract numerators $\frac{7x}{6}$

$\frac{5x}{3} + \frac{11x}{4}$

1. Same denominators $\frac{\quad}{12} + \frac{\quad}{12}$

$\frac{20x}{12} + \frac{33x}{12}$

2. Add numerators $\frac{53x}{12}$

Guided Practice

Simplify

a $\frac{y}{4} + \frac{y}{2}$

$= \frac{y}{4} + \frac{\quad}{4}$

$=$

$\frac{3y}{4}$

b $\frac{3x}{6} - \frac{x}{2}$

0

e $\frac{y}{3} - \frac{a}{4}$

$\frac{4y - 3a}{12}$

f $\frac{x}{5} + \frac{2x}{3}$

$\frac{13x}{15}$

c $3x + \frac{x}{2}$

$\frac{7x}{2}$

d $\frac{4x}{3} + \frac{3x}{10}$

$\frac{49x}{30}$

Key Ideas

1. To add or subtract fractions, the …… must be the same.
2. To make the denominators the same, find the …… of the two denominators.

1 Add and subtract these fractions.

a $\frac{a}{4} + \frac{a}{2}$

$= \frac{a}{4} + \frac{\square}{4} = \frac{\square}{\square}$

$\frac{3a}{4}$

b $\frac{r}{3} + \frac{r}{4}$

$= \frac{\square}{12} + \frac{\square}{12} = \frac{\square}{\square}$

$\frac{7r}{12}$

c $\frac{5t}{6} - \frac{2t}{3}$

$= \frac{\square}{6} - \frac{\square}{6} = \frac{\square}{\square}$

$\frac{t}{6}$

d $\frac{m}{4} + \frac{5m}{6}$

$= \frac{\square}{12} + \frac{\square}{12} = \frac{\square}{\square}$

$\frac{13m}{12}$

e $\frac{3a}{4} - \frac{2a}{5} + \frac{3a}{10}$

$= \frac{\square}{20} - \frac{\square}{20} + \frac{\square}{20} = \frac{\square}{\square}$

$\frac{13a}{20}$

f $\frac{g}{5} + \frac{2g}{3} - \frac{g}{6}$

$= \frac{\square}{30} + \frac{\square}{30} - \frac{\square}{30} = \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{7g}{10}$

2 Complete these simplifications.

a $\frac{x}{4} + \frac{2x}{3}$

$= \frac{\square}{12} + \frac{\square}{12}$

$= \frac{\square}{12}$

$\frac{11x}{12}$

b $\frac{5x}{7} - \frac{2x}{5}$

$= \frac{\square}{35} - \frac{\square}{35}$

$= \frac{\square}{35}$

$\frac{11x}{35}$

c $\frac{x+1}{2} + \frac{2x+3}{4}$

$= \frac{\square(x+1)}{4} + \frac{2x+3}{4}$

$= \frac{\square + 2x + 3}{4}$

$= \frac{\square}{4}$

$\frac{4x+5}{4}$

3 Add and subtract these fractions.

a $\frac{5t}{6} - \frac{t}{2}$

$= \frac{\square}{\square} - \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$ Simplify

$\frac{t}{3}$

b $\frac{4d}{5} + \frac{7d}{10}$

$= \frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$ Simplify

$\frac{3d}{2}$

c $\frac{n}{2} + \frac{5n}{12}$

$= \frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{11n}{12}$

d $\frac{5c}{8} - \frac{c}{3}$

$= \frac{\square}{\square} - \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{7c}{24}$

e $\frac{2q}{3} + \frac{q}{2} - \frac{3q}{4}$

$= \frac{\square}{\square} + \frac{\square}{\square} - \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{5q}{12}$

f $\frac{7e}{9} - \frac{e}{2} + \frac{5e}{6}$

$= \frac{\square}{\square} - \frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{10e}{9}$

a $\frac{4v}{5} + \frac{v}{4} - \frac{3v}{10}$

$= \frac{\square}{\square} + \frac{\square}{\square} - \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{3v}{4}$

b $\frac{3p}{4} - \frac{2p}{3} + \frac{p}{6}$

$= \frac{\square}{\square} - \frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$

$\frac{p}{4}$

4 Madeline and Abbie are working out $\frac{11a}{4} - \frac{5a}{2}$



The answer is $\frac{a}{4}$

The answer is $\frac{1}{4}a$



Explain why they are both correct.

$\frac{a}{4}$ is the same as $a \div 4 = a \times \frac{1}{4}$

4 Simplify the following. You will need to rewrite the fractions with a common denominator.

a $\frac{p}{2} - \frac{p}{3}$ $\frac{p}{6}$	**b** $\frac{a}{2} + \frac{a}{3}$ $\frac{5a}{6}$	**c** $\frac{a}{4} + \frac{a}{5}$ $\frac{9a}{20}$	**d** $\frac{p}{2} + \frac{p}{5}$ $\frac{7p}{10}$
e $\frac{q}{4} + \frac{q}{2}$ $\frac{3q}{4}$	**f** $\frac{2k}{5} + \frac{3k}{7}$ $\frac{29k}{35}$	**g** $\frac{2t}{5} - \frac{t}{3}$ $\frac{t}{15}$	**h** $\frac{9u}{11} - \frac{u}{2}$ $\frac{7u}{22}$
i $\frac{8y}{3} - \frac{5y}{6}$ $\frac{11y}{6}$	**j** $\frac{2m}{5} + \frac{4m}{3}$ $\frac{26m}{15}$	**k** $\frac{7p}{6} + \frac{2p}{5}$ $\frac{47p}{30}$	**l** $\frac{x}{4} + \frac{3x}{8}$ $\frac{5}{8}$
m $\frac{r}{3} - \frac{r}{2}$ $-\frac{r}{6}$	**n** $\frac{6u}{7} - \frac{7u}{6}$ $-\frac{13u}{42}$	**o** $\frac{4h}{9} + \frac{2d}{5}$ $\frac{38d}{45}$	**p** $\frac{5m}{12} - \frac{3m}{8}$ $\frac{m}{24}$

5 Simplify the following. Write as a single fraction. Reminder that $4x$ is the same as $\frac{4x}{1}$

a $4x + \frac{x}{3}$ $\frac{13x}{3}$	**b** $3x + \frac{x}{2}$ $\frac{7x}{2}$	**a** $\frac{a}{5} + 2a$ $\frac{11x}{5}$	**c** $\frac{8p}{3} - 2p$ $\frac{2p}{3}$
d $\frac{10u}{3} + \frac{3v}{10}$ $\frac{100u + 9v}{30}$	**b** $\frac{7y}{10} - \frac{2x}{5}$ $\frac{7y - 4x}{10}$	**c** $2t + \frac{7p}{2}$ $\frac{4t + 7p}{2}$	**e** $\frac{x}{3} - y$ $\frac{x - 3y}{3}$

6 Complete each calculation, leaving your answer as a multiple of π.

a $\pi - \frac{\pi}{3}$ $\frac{2\pi}{3}$	**b** $\pi + \frac{\pi}{3}$ $\frac{4\pi}{3}$	**c** $\pi + \frac{\pi}{4}$ $\frac{5\pi}{4}$	**d** $\pi - \frac{\pi}{4}$ $\frac{3\pi}{4}$
e $\pi + \frac{\pi}{6}$ $\frac{7\pi}{6}$	**f** $\pi - \frac{\pi}{6}$ $\frac{5\pi}{6}$	**g** $2\pi - \frac{\pi}{3}$ $\frac{t\pi}{3}$	**h** $2\pi - \frac{\pi}{4}$ $\frac{7\pi}{4}$
i $2\pi - \frac{\pi}{6}$ $\frac{11\pi}{6}$	**j** $2\pi + \frac{\pi}{3}$ $\frac{7\pi}{3}$	**k** $\pi + \frac{\pi}{2}$ $\frac{3\pi}{2}$	**l** $\pi + \frac{2\pi}{3}$ $\frac{5\pi}{3}$
m $\pi + \frac{3\pi}{4}$ $\frac{7\pi}{4}$	**n** $2\pi - \frac{3\pi}{4}$ $\frac{5\pi}{4}$	**o** $\frac{3\pi}{2} + \frac{2\pi}{3}$ $\frac{13\pi}{6}$	**p** $\frac{\pi}{3} - \frac{\pi}{4}$ $\frac{\pi}{4}$
q $\frac{7\pi}{6} - \pi$ $\frac{\pi}{6}$	**r** $\frac{4\pi}{3} - \pi$ $\frac{\pi}{3}$	**s** $\frac{5\pi}{2} - 2\pi$ $\frac{\pi}{2}$	**t** $\frac{13\pi}{6} - 2\pi$ $\frac{\pi}{6}$

7 Cedric earns an amount $\$x$ every week. He spends $\frac{1}{3}$ of his income on rent and $\frac{1}{4}$ on groceries.

a Write an algebraic fraction for the amount of money he spends on rent.

$\frac{x}{3}$

b Write an algebraic fraction for the amount of money he spends on groceries.

$\frac{x}{4}$

c Write a simplified algebraic fraction for the total amount of money he spends on rent and groceries.

$\frac{7x}{12}$

8 Edgar fills the bathtub so it is a quarter full and then adds half a bucket of water.
A full bathtub can contain T litres and a bucket contains B litres.

a Write the total amount of water in the bathtub as the sum of two algebraic fractions.

$\frac{T}{4}+\frac{B}{2}$

b Simplify the expression in part **a** to get a single algebraic fraction.

$\frac{T+2B}{4}$

9 Annie's bank account is halved in value and then \$20 is removed.
If it initially had $\$A$ in it, write an algebraic fraction for the amount left.

$\frac{A}{2}-20$ or $\frac{A-40}{2}$

10 Simplify these expressions.

a $\frac{z}{4}+\frac{z}{3}+\frac{z}{12}$

$\frac{2z}{3}$

b $\frac{2x}{5}+\frac{x}{2}-\frac{x}{5}$

$\frac{7x}{10}$

c $\frac{7u}{2}+\frac{3u}{4}-\frac{5u}{8}$

$\frac{29u}{8}$

d $\frac{8k}{3}+\frac{k}{6}-\frac{5k}{12}$

$\frac{29k}{12}$

e $\frac{p}{4}+\frac{p}{2}-3$

$\frac{3p-12}{4}$

f $\frac{5j}{12}-\frac{j}{3}+2$

$\frac{j+24}{12}$

11 Find the missing algebraic fraction. The missing fraction is fully simplified.

a $\frac{x}{2}+\frac{\square}{\square}=\frac{5x}{6}$

$\frac{2x}{3}$

b $\frac{\square}{\square}+\frac{x}{4}=\frac{3x}{8}$

$\frac{x}{8}$

c $\frac{2x}{5}+\frac{\square}{\square}=\frac{9x}{10}$

$\frac{x}{2}$

d $\frac{2x}{3}-\frac{\square}{\square}=\frac{7x}{15}$

$\frac{x}{5}$

e $\frac{\square}{\square}-\frac{x}{3}=\frac{5x}{9}$

$\frac{8x}{9}$

f $\frac{2x}{3}-\frac{\square}{\square}=\frac{5x}{12}$

$\frac{x}{4}$

Multiplying Algebraic Fractions

Review

1 Simplify algebraic multiplication.

a	$5 \times 2m$	**b**	$3 \times 6r$	**c**	$8 \times ab$
a	$2 \times -7y$	**d**	$5 \times b$	**e**	$17y \times 3$

2 Multiply fractions

a $\frac{2}{5} \times \frac{1}{2}$	**b** $\frac{2}{5} \times \frac{3}{4}$	**c** $\frac{3}{5} \times \frac{6}{7}$	**d** $\frac{2}{3} \times \frac{3}{5} \times \frac{4}{7}$

3 Simplify fractions before multiplying

a $\frac{12}{18} \times \frac{5}{7}$	**b** $\frac{8}{7} \times \frac{7}{15}$	**c** $\frac{2}{5} \times \frac{15}{24}$	**d** $\frac{8}{27} \times \frac{9}{16}$

4 Simplify algebraic fractions

a $\frac{2x}{6}$	**b** $\frac{5xy}{2y}$	**c** $\frac{7ab}{14b}$	**d** $\frac{2c}{5bc}$

Multiplying Algebraic Fractions

To multiply fractions, multiply the numerators and multiply the denominators.

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Example Multiply algebraic fractions

$$\frac{2a}{5} \times \frac{3b}{7}$$
$$= \frac{2a \times 3b}{5 \times 7}$$
$$= \frac{6ab}{35}$$

$$\frac{4x}{15} \times \frac{3x}{2}$$
$$= \frac{4x \times 3x}{15 \times 2}$$
$$= \frac{12x^2}{30}$$
$$= \frac{2x^2}{5}$$

Guided Practice

Simplify.

a $\frac{x}{3} \times \frac{2}{5}$

$\frac{2x}{15}$

b $\frac{2}{3} \times \frac{4a}{5}$

$\frac{8a}{15}$

a $\frac{6x}{5} \times \frac{7y}{6}$

$\frac{7xy}{5}$

b $\frac{8a}{5} \times \frac{3b}{4a}$

$\frac{6b}{5a}$

c $\frac{3x}{2} \times 6$

$9x$

d $\frac{10x}{3} \times \frac{6x}{5x}$

$4x$

Key Ideas

1. To multiply algebraic fractions, we multiply the …… and multiply the ……

1 Multiply these fractions.

a $\frac{x}{2} \times \frac{y}{3} = \frac{\square}{\square}$ $\frac{xy}{6}$

b $\frac{e}{5} \times \frac{k}{y} = \frac{\square}{\square}$ $\frac{ek}{5y}$

c $\frac{n}{2} \times \frac{n}{3} = \frac{\square}{\square}$ $\frac{n^2}{6}$

d $\frac{3p}{5} \times \frac{4p}{7} = \frac{\square}{\square}$ $\frac{12p^2}{35}$

e $\frac{e}{2} \times \frac{t}{3} \times \frac{5e}{4} = \frac{\square}{\square}$ $\frac{5e^2t}{24}$

f $\frac{2}{k} \times \frac{m}{3} \times \frac{4m}{5} = \frac{\square}{\square}$ $\frac{8m^2}{15k}$

2 Simplify these products.

a $\frac{x}{3} \times \frac{2}{5}$ $\frac{2x}{15}$

b $\frac{1}{7} \times \frac{a}{9}$ $\frac{a}{63}$

c $\frac{2}{3} \times \frac{4a}{5}$ $\frac{8a}{15}$

d $\frac{4c}{5} \times \frac{1}{5}$ $\frac{4c}{25}$

e $\frac{4a}{3} \times \frac{2b}{5}$ $\frac{8ab}{15}$

f $\frac{3a}{2} \times \frac{7a}{5}$ $\frac{21a^2}{10}$

3 Multiply these fractions.

a $\frac{m}{2} \times \frac{x}{m} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{x}{2}$

b $\frac{a}{3} \times \frac{6}{a} = \frac{\square}{\square} = ___$ 2

c $\frac{2c}{3} \times \frac{x}{4} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{cx}{6}$

d $\frac{ux}{2} \times \frac{x}{u} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{x^2}{2}$

e $\frac{5n}{px} \times \frac{2}{nx} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{10}{px^2}$

f $\frac{3}{k} \times \frac{2}{k} \times \frac{ak}{2} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{3a}{k}$

g $\frac{3f}{2} \times 4 \times \frac{h}{6f} = \frac{\square}{\square} = ___$ h

h $\frac{2a}{b} \times \frac{b}{2c} \times \frac{2c}{a} = \frac{\square}{\square} = ___$ 2

4 Simplify these products. Remember to simplify before you multiply.

a $\frac{6x}{5} \times \frac{7y}{6}$ $\frac{7xy}{5}$

b $\frac{2b}{5} \times \frac{7d}{6}$ $\frac{7bd}{15}$

c $\frac{8a}{5} \times \frac{3b}{4c}$ $\frac{6ab}{5c}$

d $\frac{9d}{2} \times \frac{4e}{7}$ $\frac{18de}{7}$

e $\frac{3x}{2} \times \frac{1}{6x}$ $\frac{1}{4}$

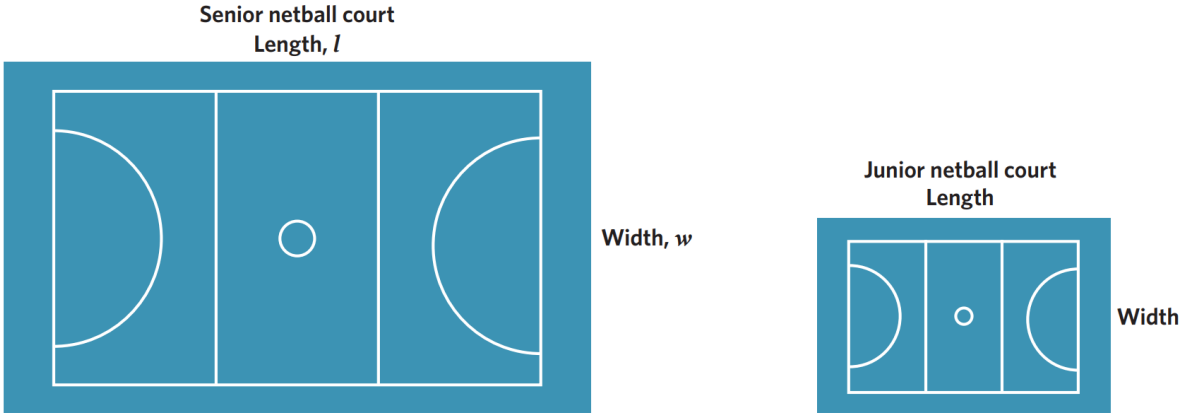
f $\frac{4}{9k} \times \frac{3k}{2}$ $\frac{2}{3}$

5 The width of a rectangle is $\frac{3x}{5}$ cm. Its length is four times its width.

Write an expression for the area of the rectangle.

$\frac{36x^2}{25}$ cm

6 The senior and junior netball courts have different dimensions.
The junior netball court has a length that is half of the senior netball court and a width that is two-thirds of the senior netball court.



a Write an expression for the area of the senior netball court.

lw

b Write an expression for the length and width of the junior netball court.

$\frac{l}{2}, \frac{2w}{3}$

c Write an expression for the area of the junior netball court.

$\frac{lw}{3}$

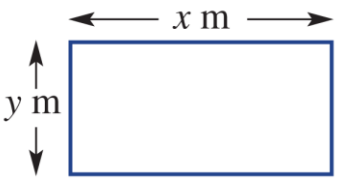
d Write an expression for the difference between the senior netball court and the junior netball court.

$\frac{2lw}{3}$

7 The rectangular field below has length x metres and breadth y metres.

a Write an expression for the area of the field.

xy



b A smaller section is fenced off. It is $\frac{1}{2}$ the length and $\frac{3}{4}$ the breadth.

i Write an expression for the length and breadth of the smaller section.

$\frac{x}{2}, \frac{3y}{4}$

ii Hence, write an expression for the area of the smaller section.

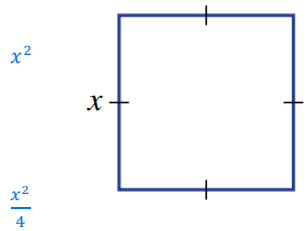
$\frac{3xy}{8}$

c Find the percentage of the smaller area compared to the larger area.

37.5%

8 Consider a square with side length x.

a Write an expression for the area of the square.



b The length of each side is now halved.
Give an expression for the area of the new square.

c If each side of the original square is multiplied by 0.7, find an expression for the area of the square, as an algebraic fraction. Recall that $0.7 = \frac{7}{10}$

$\frac{49x^2}{100}$

d If each side of the original square is three-fifths of the original length, would the new area be more or less than half of the original area? Show your reasoning.

The new area is $\frac{9x^2}{25}$, which is less than $\frac{1}{2}$ of x^2

e If you want to halve the area of the original square, by what factor would you have to divide each length?

Divide by $\sqrt{2}$

f If you continued to reduce each side of the original square by a factor of $\frac{3}{5}$, show that the area after 2 reductions is $\frac{81}{625}x^2$

g If you continued to reduce each side of the original square by a factor of $\frac{3}{5}$, write an expression for the area after n reductions.

$\left(\frac{3}{5}\right)^{2n} x^2$

Dividing Algebraic Fractions

Review

1 Identify the reciprocal.

a $\frac{2}{3}$

b $\frac{7}{8}$

c 12

d $\frac{1}{5x}$

2 Divide fractions.

a $\frac{2}{3} \div \frac{4}{5}$

b $\frac{1}{2} \div \frac{1}{4}$

c $\frac{3}{4} \div \frac{2}{5}$

d $\frac{1}{3} \div 4$

Rule for Dividing Fractions

To divide by a fraction, multiply by the reciprocal.

$$\frac{a}{b} \div \frac{c}{d}$$
$$= \frac{a}{b} \times \frac{d}{c}$$

Example Divide algebraic fractions

$$\frac{3a}{8} \div \frac{b}{5}$$
$$= \frac{3a}{8} \times \frac{5}{b}$$
$$= \frac{15a}{8b}$$

$$\frac{u}{4} \div \frac{15p}{2}$$
$$= \frac{u}{4} \times \frac{2}{15p}$$
$$= \frac{2u}{60p} = \frac{u}{30p}$$

Guided Practice

Simplify.

a $\frac{3a}{5} \div \frac{1}{5}$

$3a$

b $\frac{9a}{10} \div \frac{1}{4}$

$\frac{18a}{5}$

e $\frac{4}{5} \div \frac{2y}{3}$

$\frac{6}{5y}$

f $\frac{4a}{7} \div \frac{2}{5}$

$\frac{10a}{7}$

g $\frac{4x}{5} \div 3$

$\frac{4x}{15}$

h $2 \div \frac{x}{5}$

$\frac{10}{x}$

Key Ideas

1. Dividing by a fraction is the same as multiplying by the ……

1 Divide these fractions.

a $\frac{c}{2} \div \frac{3}{v} = \frac{c}{2} \times \frac{v}{3} = \frac{cv}{6}$ (÷ → ×; reciprocal)

b $\frac{a}{c} \div \frac{d}{b} = \square \times \square = \square$ $\frac{ab}{cd}$

c $\frac{2}{u} \div \frac{u}{2} = \square \times \square = \square$ $\frac{4}{u^2}$

d $\frac{d}{3} \div \frac{4}{d} = \square \times \square = \square$ $\frac{d^2}{12}$

e $\frac{y}{3} \div \frac{1}{2} = \square \times \square = \square$ $\frac{2y}{3}$

f $\frac{2n}{3} \div \frac{5}{x} = \square \times \square = \square$ $\frac{2nx}{15}$

g $\frac{3c}{m} \div 5 = \square \times \square = \square$ $\frac{3c}{5m}$

h $\frac{4k}{3} \div \frac{8}{3} = \square \times \square = \square = \square$ $\frac{k}{2}$

i $\frac{6y}{5} \div \frac{3}{4g} = \square \times \square = \square = \square$ $\frac{8gy}{5}$

2 Simplify these divisions. Remember to simplify before you multiply.

a $\frac{2}{3} \div \frac{4x}{7}$ $\frac{7}{6x}$

b $\frac{4}{5} \div \frac{2y}{3}$ $\frac{6}{5y}$

c $\frac{1}{7} \div \frac{2}{x}$ $\frac{x}{14}$

d $\frac{4a}{7} \div \frac{2}{5}$ $\frac{10x}{7}$

e $\frac{4b}{7} \div \frac{2c}{5}$ $\frac{10}{7c}$

f $\frac{2x}{5} \div \frac{4y}{3}$ $\frac{3}{10y}$

g $\frac{2y}{x} \div \frac{3}{y}$ $\frac{2y^2}{3x}$

h $\frac{5}{12x} \div \frac{7x}{2}$ $\frac{5}{42x^2}$

i $\frac{4a}{5} \div \frac{2b}{7a}$ $\frac{14a}{5b}$

3 Simplify the following.

a $3 \times \frac{4x}{5}$ $\frac{12x}{5}$

b $\frac{4x}{5} \div 3$ $\frac{4x}{15}$

c $2 \div \frac{x}{5}$ $\frac{10}{x}$

d $4 \times \frac{a}{3}$ $\frac{4a}{3}$

e $5 \times \frac{7}{10x}$ $\frac{7}{2x}$

f $1 \div \frac{x}{2}$ $\frac{2}{x}$

4 Simplify these expressions. Only invert the fractions being divided.

a $\frac{k}{3} \div \frac{2}{3} \times \frac{4}{k}$

$\frac{\square}{\square} \times \frac{\square}{\square} \times \frac{\square}{\square} = \frac{\square}{\square} = __$ 2

b $\frac{4}{e} \times \frac{2e}{3} \div \frac{1}{e}$

$\frac{\square}{\square} \times \frac{\square}{\square} \times \square = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{8e}{3}$

c $\frac{ch}{2} \div \frac{1}{2} \times \frac{a}{h}$

$\frac{\square}{\square} \times \square \times \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$ ac

d $\frac{2v}{5} \times \frac{3}{2} \div \frac{v^2}{3}$

$\frac{\square}{\square} \times \frac{\square}{\square} \times \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{9}{5v}$

e $\frac{1}{d} \times \frac{3d}{4} \div \frac{1}{d}$

$\frac{\square}{\square} \times \frac{\square}{\square} \times \square = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{3d}{4}$

f $\frac{q}{p} \times \frac{aq}{2p} \div \frac{4q}{5p}$

$\frac{\square}{\square} \times \frac{\square}{\square} \times \frac{\square}{\square} = \frac{\square}{\square} = \frac{\square}{\square}$ $\frac{5aq}{8p}$

5 Krystal's family goes to dinner with Helen's family.
The bill comes to a total of $\$x$ and each family pays half.

a Write an algebraic fraction for the amount Krystal's family pays.

$\frac{x}{2}$

b Krystal says she will pay one third of her family's bill.
Write an algebraic fraction for the amount that she pays.

$\frac{x}{6}$

6 Write an algebraic fraction for the result of the following operations.

a A number q is halved and then the result is tripled.

$\frac{3q}{2}$

b A number x is multiplied by $\frac{2}{3}$ and the result is divided by $1\frac{1}{3}$.

$\frac{x}{2}$

c A fraction $\frac{a}{b}$ is multiplied by its reciprocal.

1

d The number x is reduced by 25% and then halved.

$\frac{3x}{8}$

7 Simplify the expressions.

a

$$5mn \times \frac{4d}{n} \times \frac{1}{15mn}$$

$\frac{3d}{3n}$

b

$$\frac{6}{r} \times \frac{5r}{9} \div \frac{15}{xy}$$

$\frac{2xy}{9}$

c

$$\frac{6cf}{q} \div \frac{5}{qf} \div \frac{10c}{a}$$

$\frac{3a}{25}$

d

$$\frac{4ab}{5} \div \frac{a}{2b} \div \frac{8b}{15}$$

$3b$

8 Expand and simplify the following.

a

$$\frac{1}{2}(x+4)$$

$\frac{x}{2}+2$

b

$$\frac{2}{3}(y-3)$$

$\frac{2y}{3}-2$

c

$$\frac{3}{4}(2a+4b)$$

$\frac{3a}{2}+3b$

d

$$\frac{1}{2}(6x+8)+\frac{1}{3}(12x-3)$$

$7x+3$

e

$$\frac{2}{3}(12y-6)+\frac{1}{4}(2y+3)$$

$\frac{17y}{2}-\frac{13}{4}$

f

$$\frac{x}{3}(6+9y)+y(x+4)$$

$2x+4xy+4y$